



CHE 109: Engineering Chemistry – I
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Basics of Chemical Hazard and
Chemical Weapons Convention (CWC)
Chapter 11

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What does "hazardous" mean?

Any chemical that can potentially cause harm is considered to be hazardous. Chemicals can be hazardous if they:

- Can irritate the eyes, skin, or respiratory tract
- Are toxic or carcinogenic (cancer-causing)
- Are corrosive
- Can explode or cause fire
- Create low-oxygen environments (e.g., by displacing or chemically reacting with oxygen in an enclosed or confined space)

GHS Pictogram	Characteristics	Storage Considerations
Explosives 	Capable of self-sustained chemical reaction, of producing gas at such temperature, pressure speed as to damage surroundings; or is manufactured for the purpose of producing explosion.	Dry. Cool area protected from shock, rising temperature changes. Site should be separate from other storage/activities. Magazines should be heavily constructed.
Flammable Gas 	A compressed gas and forms a flammable mixture at normal atmospheric pressure over a significant range of concentration.	Ventilated, cool, dry area away from heat and ignition sources. Cylinder temperature should never exceed 51°C (125°F). Stored secure in an upright position and care taken to avoid damaging valves. Separate full and empty containers.
Non Flammable Gas 	A compressed gas, dissolved gas, or liquefied by compression or refrigeration.	Ventilated, cool, dry area away from heat and ignition sources. Avoid elevation of cylinder temperature and store secured in an upright position. Avoid damaging valves. Separate full and empty containers.
Dangerously Reactive When Wet 	When in contact with water or water vapour emits a flammable or toxic gas; becomes spontaneously combustible under conditions of shock or increased temperatures or pressure or can undergo hazardous decompositions.	Ventilated, cool, dry and waterproof area. Ensure no water sprinkler system in area and use alternative fire fighting system (i.e. Halon suppression). Containers should be elevated above potential flood levels.
Oxidizers 	Liable to cause or contribute to the combustion of other materials by yielding oxygen or other oxygen material; or is an organic peroxide.	Ventilated, cool, dry area away from direct sunlight. Protect from temperature elevation or rapid temperature changes. Fire fighting procedures should be reviewed as normal.
Flammable/Combustible Liquids and Solids 	Can be ignited at normal temperatures; or can cause a fire through friction or retained heat.	Ventilated, cool, dry area. Eliminate possible ignition sources. Ensure proper firefighting equipment. Storage area should be fireproofed and comply with fire code.
Acids 	Can significantly corrode steel or aluminum; or is corrosive to skin and has a pH $\leq$ 2.0.	Ventilated, cool, dry area away from direct sunlight in approved containers. Protect storage area from rapid temperature changes. Structural and stacking materials should not be non-corroding or metal covered with acid fume resistant paint.
Bases 	Can significantly corrode steel or aluminum; or is corrosive to skin and has a pH $\geq$ 12.5.	Ventilated, cool, dry area away from direct sunlight in approved containers. Protect storage area from temperature elevation and rapid changes. Structural and stacking materials should not be non-corroding or metal covered with acid fume resistant paint.
Toxins 	Is poisonous, and or radioactive, and/or infectious, and/or hazardous to the environment, and waste materials may be hazardous.	Ensure containers are secure, closed and properly labelled.

Chemical weapons are toxic substances used to inflict harm or death by releasing chemicals that interfere with normal bodily functions. They are classified based on their chemical properties, effects on the body, and intended use. Here's an overview of the classification of chemical weapons:

1. Nerve Agents

Nerve agents are highly toxic chemicals that disrupt the nervous system, preventing the body from controlling essential functions such as breathing. They are considered the most deadly form of chemical warfare.

•Examples:

- Sarin (GB)
- VX
- Tabun (GA)
- Soman (GD)

•Symptoms: Drooling, difficulty breathing, convulsions, paralysis, and death.

2. Blister Agents (Vesicants)

Blister agents cause severe chemical burns and blisters on the skin and mucous membranes. These agents are often used to incapacitate the enemy.

•Examples:

- Mustard gas (Sulfur mustard)
- Lewisite
- Phosgene oxime

•Symptoms: Painful blisters, eye irritation, coughing, and difficulty breathing. In severe cases, systemic poisoning can occur, leading to death.

3. Choking Agents (Pulmonary Agents)

Choking agents target the respiratory system and can cause suffocation by damaging the lungs and airways.

•**Examples:**

- Chlorine gas
- Phosgene

•**Symptoms:** Coughing, difficulty breathing, chest pain, and in severe cases, death due to asphyxiation.

4. Blood Agents

Blood agents interfere with the body's ability to use oxygen by affecting the hemoglobin or enzymes in red blood cells.

•**Examples:**

- Cyanide (Hydrogen cyanide, Cyanogen chloride)
- Arsine

•**Symptoms:** Headache, dizziness, confusion, difficulty breathing, and loss of consciousness. High doses can cause death within minutes.

5. Riot Control Agents (Tear Gas and Vomiting Agents)

Riot control agents are used to disperse crowds or incapacitate individuals temporarily. Though not classified as lethal chemical weapons, they are still considered chemical agents.

•**Examples:**

- Tear gas (CS gas)
- Chloropicrin
- Oleoresin capsicum (OC, commonly known as pepper spray)

- **Symptoms:** Eye watering, coughing, sneezing, nausea, and vomiting.

6. Incapacitating Agents

These chemicals are designed to incapacitate or impair individuals without necessarily causing death.

•Examples:

- BZ (3-Quinuclidinyl benzilate)

•**Symptoms:** Drowsiness, confusion, hallucinations, loss of coordination, and potential long-term psychological effects.

Legal Framework

The use of chemical weapons is prohibited under international law, notably by the **Chemical Weapons Convention (CWC)**, which came into force in 1997. The treaty aims to eliminate the production, stockpiling, and use of chemical weapons and to promote peaceful uses of chemistry. Violations of the CWC are considered war crimes.

Each of these classes of chemical weapons is designed to achieve specific military goals, from immediate lethality to prolonged incapacitation or disruption of the enemy's capabilities.